1 Lie Algebra

1.1 Definitions and Structure of Lie Algebra

Definition 1. *A vector space L over a field F with an operation* $\begin{array}{ccc} L \times L & \to & L \\ (x,y) & \mapsto & [xy] \end{array}$ *called the bracket of x, y is called a Lie algebra over F if the following axioms are satisfied;* $\left\{ \begin{array}{ll} i. & \text{the bracket is bilinear} \\ ii. & [xx]=0,\ \forall x \in L \\ iii. & [x[yz]]+[y[zx]]+[z[xy]]=0,\ \forall x,y,z \in L \end{array} \right.$.

$$\rightarrow \begin{array}{rl} [x+y,x+y] = [xx]+[xy]+[yx]+[yy] & \text{by the bilinearity } i \\ = [xy]+[yx]=0 & \text{by } ii \end{array} \rightarrow [xy]=-[yx].$$

Definition 2. *We say two Lie algebras L, L' over F are isomorphic if there exists a vector space isomorphism $\phi : L \to L'$ s.t. $\phi([xy]) = [\phi(x)\phi(y)],\ \forall x,y \in L$.*

Definition 3. *The Lie-subalgebra of L is a subspace K of L s.t. $[xy] \in K,\ \forall x,y \in K$; in particular, K is a Lie algebra in its own right relative to the inherited operations.*

If V is a finite dimensional vector space over F, the set of linear transformations $V \to V$ is denoted by End$V$. It is a ring relative to the usual product operation. Then for the bracket $[xy] = xy - yx$, End$V$ becomes a Lie algebra over F. Write $\mathfrak{gl}(V)$ for End$V$ viewed as Lie algebra and call it the general linear algebra. Any subalgebra of a Lie algebra $\mathfrak{gl}(V)$ is called a linear Lie algebra.

e.g.

- $\mathfrak{sl}(V)$ is the set of endomorphisms of V having trace zero, called **the special linear algebra**. It is a proper subalgebra of $\mathfrak{gl}(V)$, hence of dimension at most $(l+1)^2-1$, while the number of linearly independent of matrices of trace zero is $l+(l+1)^2-(l+1)$ matrices.
- When $\dim V = 2l$ with basis $(v_1,\ldots,v_{2l})$. $\mathfrak{sp}(V)$ is the set of all endomorphisms x of V satisfying $f(x(v),w) = -f(v,x(w))$, called the symplectic algebra.

  When $\dim V = 2l+1$, $\mathfrak{o}(V)$ is the set of all endomorphisms x of V satisfying $f(x(v),w) = -f(v,x(w))$, called the orthogonal algebra.
- $\left\{ \begin{array}{ll} \mathfrak{t}(n,F) & \text{is the set of upper triangular matrices } (a_{ij}),\ a_{ij}=0,\ \forall i>j \\ \mathfrak{d}(n,F) & \text{is the set of all diagonal matrices.} \\ \mathfrak{n}(n,F) & \text{is the set of all strictly upper triangular matrices } (a_{ij}),\ a_{ij}=0,\ \forall i \geq j \end{array} \right.$

  $\rightarrow \mathfrak{t}(n,F) = \mathfrak{d}(n,F) + \mathfrak{n}(n,F)$ with $[\mathfrak{d}(n,F), \mathfrak{n}(n,F)] = \mathfrak{n}(n,F)$.

By an F-algebra, we mean a vector space $\mathfrak{U}$ over F endowed with a bilinear operation $\mathfrak{U} \times \mathfrak{U} \to \mathfrak{U}$, usually denoted by juxtaposition.

Definition 4. *A derivation of the F-algebra $\mathfrak{U}$ is a linear map $\delta : \mathfrak{U} \to \mathfrak{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$. The collection of all derivations of $\mathfrak{U}$ is denoted by Der$\mathfrak{U}$, which is a vector subspace of End$\mathfrak{U}$.*

$\rightarrow$ Der$\mathfrak{U}$ is a subalgebra of $\mathfrak{gl}(\mathfrak{U})$ ($x,y \in \mathrm{Der}(\mathfrak{U})$, $[xy](ab) = (xy-yx)(ab) = (xy)(ab) - (yx)(ab) = (x)(ay(b)+y(a)b) - (y)(ax(b)+x(a)b) = a(xy)(b) + x(a)y(b) + y(a)x(b) + (xy)(a)b - a(yx)(b) - y(a)x(b) - x(a)y(b) - (yx)(a)b = a(xy)(b) + (xy)(a)b - a(yx)(b) - (yx)(a)b$, i.e. $xy \in \mathrm{Der}(\mathfrak{U})$). Since a Lie algebra L is an F-algebra in the above sense, Der$L$ is defined. If $x \in L, y \mapsto [xy]$ is an endomorphism of L, which we denote ad$x \in$ Der$L$. Derivations of this form are called **inner**, all others **outer**.

Definition 5. *The map* $\begin{array}{ccc} L & \to & DerL \\ x & \mapsto & adx \end{array}$ *is called the adjoint representation of L.*

If L is any Lie algebra with basis $x_1,\ldots,x_n$, the entire multiplication table of L can be recovered from the structure constants a_{ij}^k which occur in the expressions $[x_ix_j] = \sum_{k=1}^n a_{ij}^k x_k$, where $\left\{ \begin{array}{rl} a_{ii}^k = 0 = a_{ij}^k + a_{ji}^k & \text{by } ii \\ \sum_k (a_{ij}^k a_{kl}^m + a_{jl}^k a_{ki}^m + a_{li}^k a_{kj}^m) = 0 & \text{by } iii \end{array} \right.$

Definition 6. *A subspace I of a Lie algebra L is called an ideal of L if $x \in L, y \in I$ imply $[xy] \in I$.*

e.g.

- 0 and L itself are ideals
- $\left\{ \begin{array}{rl} \text{the centre} & Z(L) = \{z \in L \mid [xz] = 0, \ \forall x \in L\} \\ \text{the derived algebra} & [LL] \end{array} \right.$
- for I, J ideals, $I + J$, $[IJ]$ are ideals.

Definition 7. *If L has no ideals except itself and 0, and if $[LL] \neq 0$, we call L simple.*

The quotient algebra L/I is formally the same as the construction of a quotient ring; as vector space L/I is just the quotient space, while its Lie multiplication is defined by $[x + I, y + I] = [xy] + I$. **The normaliser** of a subalgebra K of L is defined by $N_L(K) = \{x \in L \mid [xK] \subset K\}$. Then by the Jacobi identity, $N_L(K)$ is a subalgebra of L. If $K = N_L(K)$, it is called **self-normalising**. **The centraliser** $C_L(X) = \{x \in L \mid [xX] = 0\}$ is a subalgebra of L by the Jacobi identity.

Definition 8. *Given Lie algebras L, L' over F, a linear transformation $\phi : L \to L'$ is called a homomorphism if $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$. ϕ is called a monomorphism if $\ker \phi = 0$, an epimorphism if $Im\phi = L'$, an isomorphism if it is both monomorphism and epimorphism.*

$$\to \left\{ \begin{array}{ll} \ker \phi \text{ is an ideal of } L & (\forall x \in \ker \phi, \quad \forall y \in L.\ \phi([xy]) = [\phi(x)\phi(y)] = 0) \\ \mathrm{Im}\phi \text{ is a subalgebra of } L' & \end{array} \right. .$$

Proposition 1.

1. *If $\phi : L \to L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong Im\phi$. If I is any ideal of L included in $\ker \phi$, there exists a unique homomorphism $\psi : L/I \to L'$ making*
$$\begin{tikzcd} L \arrow[r, "\phi"] \arrow[dr, "\pi"] & L' \\ & L/I \arrow[u, "\psi"'] \end{tikzcd}$$
commutes.
2. *If I, J are ideals of L s.t. $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J.*
3. *If I, J are ideals of L, there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.*

$\to$ same as the standard homomorphism theorems.

Definition 9. *A representation of a Lie algebra L is a homomorphism $\phi : L \to \mathfrak{gl}(V)$, where V is a vector space over a field F.*

e.g. adjoint representation $\begin{array}{rcc} \mathrm{ad} : L & \to & \mathfrak{gl}(L) \\ x & \mapsto & \mathrm{ad}x \end{array}$ preserves the bracket ($[\mathrm{ad}x, \mathrm{ad}y](z) = \mathrm{ad}x\mathrm{ad}y(z) - \mathrm{ad}y\mathrm{ad}x(z) = \mathrm{ad}x([yz]) - \mathrm{ad}y([xz]) = [x[yz]] - [y[xz]] = [x[yz]] + [[xz]y] = [[xy]z] = \mathrm{ad}[xy](z)$). The kernel of ad consists of all $x \in L$ s.t. $[xy] = 0$, $\forall y \in L$ i.e. kerad$=Z(L)$.

Definition 10. *An automorphism of L is an isomorphism of L onto itself and $AutL$ denotes the group of all such.*

For char$F = 0$, and ad$x$ nilpotent, $\exp(\mathbf{ad}x) \in \mathbf{Aut}L$.

pf. Let δ be any nilpotent derivation of L. $\exp \delta(x) \exp \delta(y) = (\sum_{i=0}^{k-1} (\frac{\delta^i x}{i!}))(\sum_{j=0}^{k-1} (\frac{\delta^j y}{j!})) = \sum_{n=0}^{2k-2} (\sum_{i=0}^{n} (\frac{\delta^i x}{i!})(\frac{\delta^{n-i} y}{(n-i)!})) = \sum \frac{\delta^n (xy)}{n!} = \sum_{n=0}^{k-1} \frac{\delta^n (xy)}{n!} = \exp \delta(xy)$. Since $-\delta$ is also a nilpotent derivation, $\exp(-\delta)$ is defined and $\exp \delta \exp(-\delta) = \mathrm{id}$, so $\exp \delta \in \mathrm{Aut}L$.

$\to$ an automorphism of the form $\exp \mathrm{ad}x$, where ad$x$ nilpotent, is called **inner**. More generally, the subgroup of Aut$L$ generated by these is denoted Int$L$ and its elements are called inner automorphisms. Then it is a normal subgroup ($\forall \phi \in \mathrm{Aut}L$, $\forall x \in L$, $\phi(\mathrm{ad}x)\phi^{-1} = \mathrm{ad}\phi(x)$ so $\phi(\exp \mathrm{ad}x)\phi^{-1} = \exp \mathrm{ad}\phi(x)$).

Definition 11. *Define a sequence of ideals of L by $L^{(0)} = L$, $L^{(1)} = [LL]$, $L^{(2)} = [L^{(1)}L^{(1)}], \ldots, L^{(i)} = [L^{(i-1)}L^{(i-1)}]$, called the derived series. Then L is solvable if $L^{(n)} = 0$ for some n.*

$\mathfrak{t}(n, F)$ **is solvable.**
pf. The obvious basis for $L = \mathfrak{t}(n, F)$ consists of the matrix units e_{ij} for which $i \leq j$ with its dimension $n(n+1)/2$. Here $[e_{ii}, e_{il}] = e_{il}$, $\forall i < l \rightarrow \mathfrak{n}(n, F) \subset [LL]$ where $\mathfrak{n}(n, F)$ is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$ and $\mathfrak{d}(n, F)$ is abelian, $\mathfrak{n}(n, F)$ is equal to the derived algebra of L. Then inside the algebra $\mathfrak{n}(n, F)$, we set the level for e_{ij} as $j - i$. Assume $i < j, k, l$. Then $[e_{ij}, e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. In particular, each e_{il} is commutator of two matrices whose levels add up to that of e_{il}. So $L^{(2)}$ is spanned by those e_{ij} of level ≥ 2, $L^{(i)}$ by those of level $\geq 2^{i-1}$. Finally whenever $2^{i-1} > n - 1$, $L^{(i)} = 0$ and the stmt follows.

Proposition 2. *Let L be a Lie algebra.*

1. *If L is solvable, then so are all subalgebras and homomorphic image of L.*
2. *If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable*
3. *If I, J are solvable ideals of L, then so is $I + J$.*

pf.

- by the definition of the subalgebra, if K is a subalgebra of L, $K^{(i)} \subset L^{(i)}$. Similarly, if $\phi : L \rightarrow M$ is an epimorphism, $\phi(L^{(i)}) = \phi(L) = M = M^{(i)}$ when $i = 0$. For $i > 0$, $\phi([L^{(i-1)}L^{(i-1)}]) = [\phi(L^{(i-1)})\phi(L^{(i-1)})] = [M^{(i-1)}M^{(i-1)}] = M^{(i)}$ by the inductive hypothesis. Thus the proposition follows.
- Say $(L/I)^{(n)} = 0$. Then by applying 1 to the canonical homomorphism $\pi : L \rightarrow L/I$, $\pi(L^{(n)}) = 0$ or $L^{(n)} \subset I = \ker \pi$. If $I^{(m)} = 0$, $(L^{(i)})^{(j)} = L^{(i+j)}$ implies $L^{(n+m)} = 0$.
- As a homomorphic image of I, $I/(I \cap J)$ is solvable by 1. Then by the previous proposition 3, the isomorphic one $(I + J)/J$ is also solvable. Then so is $I + J$ by 2 applied to the pair $I + J$ and J.

Let L be an arbitrary Lie algebra and let S be a maximal solvable ideal. If I is any other solvable ideal of L, then by the previous proposition 3, $S + I = S$ by maximality or $I \subset S$. Thus there exists a unique maximal solvable ideal, called the **radical** of L, denoted Rad$L$. When Rad$L = 0$, L is called semisimple.

Definition 12. *Define a sequence of ideals of L by $L^0 = L$, $L^1 = [LL]$, $L^2 = [LL^1], \ldots, L^i = [LL^{i-1}]$, called the descending central series.*

$\rightarrow L$ is called nilpotent if $L^n = 0$ for some n.

Proposition 3. *Let L be a Lie algebra.*

1. *If L is nilpotent, then so are all subalgebras and homomorphic images of L*
2. *If $L/Z(L)$ is nilpotent, then so is L.*
3. *If L is nilpotent and nonzero, then $Z(L) \neq 0$.*

pf.

- similarly with the previous proposition 1
- $L^n \subset Z(L)$ implies $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$ so the stmt follows.
- The last nonzero term of the descending central series is central by the definition.

$\rightarrow \exists n$ s.t. $\mathrm{ad}x_1 \mathrm{ad}x_2 \ldots \mathrm{ad}x_n(y) = 0$, $\forall x_i, y \in L$ implies L is nilpotent. If L is any Lie algebra and $x \in L$, we call x **ad-nilpotent** if ad$x$ is a nilpotent endomorphism. If L is nilpotent then all elements of L are ad-nilpotent.

Lemma 1. *Let $x \in \mathfrak{gl}(V)$ be a nilpotent endomorphism. Then ad$x$ is also nilpotent.*

pf. Associate to x two endomorphisms, $\begin{cases} \lambda_x(y) = xy & \text{left translation} \\ \rho_x(y) = yx & \text{right translation} \end{cases}$, which are nilpotent because x is. Moreover λ_x and ρ_x commute trivially. In any ring the sum or difference of two commuting nilpotents is nilpotent (x, y commuting nilpotents. then $(x-y)^n = x^n + \binom{n}{1}x^{n-1}(-y) + \cdots + (-y)^n = 0$ when $n > 2\max(n_1, n_2)$) so ad$x = \lambda_x - \rho_x$ is nilpotent.

Theorem 1. *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, then there exists nonzero $v \in V$ for which $Lv = 0$.*

pf. When $n \equiv \dim L = 0$ or 1, the theorem holds trivially.
Suppose $K \subsetneq L$ be any proper subalgebra of L. By the previous lemma, K acts via ad as a Lie algebra of nilpotent linear transformations on the vector space L, hence also on the vector space L/K. Because $\dim K < \dim L$, by the inductive hypothesis, $\exists$ a vector $x + K \neq K$ in L/K killed by the image of K in $\mathfrak{gl}(L/K)$. This means $[yx] \in K$, $\forall y \in K, x \notin K$. In other words, K is properly included in $N_L(K)$. Take K to be a maximal proper subalgebra of L. Then $N_L(K)$ should be L, i.e. K is an ideal of L. If $\dim L/K$ were greater than 1, the inverse image in L of one dimensional subalgebra of L/K would be a proper subalgebra containing K, a contradiction to the maximality. Thus K has codimension 1. Thus $L = K + Fz$, $z \in L \setminus K$. By induction $W = \{v \in V \mid Kv = 0\}$ is nonzero. Since K is an ideal, W is stable under L. Choose $z \in L \setminus K$, so the nilpotent endomorphism z has an eigenvector, i.e. $\exists 0 \neq v \in W$ for which $zv = 0$. Finally $Lv = 0$ as desired.

Theorem 2 (Engel)**.** *If all elements of L are ad-nilpotent, then L is nilpotent.*

pf. We are given a Lie algebra L all of whose elements are ad-nilpotent. Thus the algebra ad$L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus there exists $0 \neq x \in L$ for which $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ evidently consists of ad-nilpotent elements and has smaller dimension than L. By the inductive hypothesis, $L/Z(L)$ is nilpotent. By the previous proposition 2, L itself is nilpotent.

Corollary 1. *Under the hypotheses of the theorem, there exists a flag (V_i) in V stable under L with $xV_i \subset V_{i-1}$, $\forall i$. In other words, there exists a basis of V relative to which the matrices of L are all in $\mathfrak{n}(n, F)$.*

pf. Begin with any nonzero $v \in V$ killed by L, which is possible by the previous theorem. Set $V_1 = Fv$. Let $W = V/V_1$. Then the induced action of L on W is also by nilpotent endomorphisms. By the inductive hypotheses, W has a flag stabilised by L, whose inverse image in V does the trick.

Lemma 2. *Let L be nilpotent, K an ideal of L. Then if $K \neq 0$, $K \cap Z(L) \neq 0$.*

pf. L acts on K via the adjoint representation, so by the previous theorem yields nonzero $x \in K$ killed by L, i.e. $[Lx] = 0$, so $x \in K \cap Z(L)$.

Theorem 3. *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If $V \neq 0$, then V contains a common eigenvector for all the endomorphisms in L.*

pf. When $\dim L = 0$, the theorem holds trivially. Let $\dim L > 0$. Since L is solvable of positive dimension, L properly includes $[LL]$. Since $L/[LL]$ is abelian, any subspace is automatically an ideal. Take a subspace of codimension 1, then its inverse image K is an ideal of codimension 1 in L including $[LL]$. By the inductive hypothesis, $\exists v \in V$ for K. This means $\forall x \in K$, $xv = \lambda(x)v$, where $\lambda : K \to F$ is some linear function. Fix this λ and denote by $W = \{w \in V \mid xw = \lambda(x)w, \ \forall x \in K\}$. Then $W \neq 0$. Let $w \in W, x \in L$ be fixed and n be the smallest integer for which $w, xw, \ldots, x^n w$ are LI. Let W_i be the subspace of V spanned by $w, xw, \ldots, x^{i-1}w$, so $\dim W_n = n$, $W_n = W_{n+i}$ and x maps W_n into W_n. Then $\forall y \in K$ leaves each W_i invariant. Relative to the basis $w, xw, \ldots, x^{n-1}w$ of W_n, $yx^i w \equiv \lambda(y)x^i w \mod W_i$, $y \in K$ is represented by an upper triangular matrix whose diagonal entries equal $\lambda(y)$. Then $\mathrm{T}_{W_n}(y) = n\lambda(y)$. In particular this is true for elements of K of the special form $[xy]$. But x, y both stabilise W_n so $[xy]$ acts on W_n as the commutator of the two endomorphisms of W_n and therefore its trace is 0. So $n\lambda([xy]) = 0 \to \lambda([xy]) = 0$, given char $F = 0$. Take an arbitrary $y \in K$. Then $yxw = xyw - [xy]w = \lambda(y)xw - \lambda([xy])w = \lambda(y)xw$. So L stabilise W and L leaves W invariant. Write $L = K + Fz$, and given the fact that F is algebraically closed, we can find an eigenvector $v_0 \in W$ of z. Then v_0 is a common eigenvector for L and the theorem follows by induction.

Corollary 2 (Lie's Theorem)**.** *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then L stabilises some flag in V, i.e. the matrices of L relative to a suitable basis of V are upper triangular.*

pf. By the induction on $\dim V$ with the previous theorem.

Corollary 3. *Let L be solvable. Then there exists a chain of ideals of L, $0 = L_0 \subset L_1 \subset \cdots \subset L_n = L$ s.t.* $\dim L_i = i$.

Corollary 4. *Let L be solvable. Then $x \in [LL]$ implies $ad_L x$ is nilpotent. In particular,$[LL]$ is nilpotent.*

pf. Find a flag of ideals by the previous corollary. Relative to a basis $(x_1, \ldots, x_n)$ of L for which $(x_1, \ldots, x_i)$ spans L_i, the matrices of adL lie in $\mathfrak{t}(n, F)$. Thus the matrices of $[adL, adL] = ad_L[LL]$ lie in $\mathfrak{n}(n, F)$, the derived algebra of $\mathfrak{t}(n, F)$. Thus $ad_L x$ is nilpotent for $x \in [LL]$. Since $ad_{[LL]}x$ is nilpotent, so is $[LL]$ by the Engel's theorem.

The Jordan canonical form for a single endomorphism x over an algebraically closed field amounts to an expression of x in matrix form as a sum of blocks $\begin{bmatrix} a & 1 & & & 0 \\ & a & 1 & & \\ & & \cdot & \cdot & \\ & & & \cdot & 1 \\ 0 & & & & a \end{bmatrix}$. Since $diag(a, \ldots, a)$ commutes with the nilpotent matrix having one's just above the diagonal and zeros elsewhere, x is the sum of a diagonal and a nilpotent matrix which commute.

Call $x \in \mathrm{End}V$ semisimple, if the roots of its minimal polynomial over F are all distinct. Equivalently x is semisimple iff x is diagonalisable.

Proposition 4. *Let V be a finite dimensional vector space over F, $x \in EndV$.*

1. *$\exists! x_s, x_n \in EndV$ satisfying the conditions $x = x_s + x_n$ where x_s is semisimple, x_n is nilpotent, and x_s, x_n commute.*
2. *$\exists p(T), q(T)$ polynomials in one indeterminate, without constant term s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, x_s, x_n commute with any endomorphism commuting with x.*
3. *If $A \subset B \subset V$ are subspaces, and x maps B into A, then x_s, x_n also map B into A.*

pf.

- 2

  Let $a_1, \ldots, a_k$ be the distinct eigenvalues of x, so the characteristic polynomial is $\Pi(T - a_i)^{m_i}$ where m_i is the multiplicity of a_i. If $V_i = \ker(x - a_i \cdot 1)^{m_i}$, then V is the direct sum of the subspaces $V_1, \ldots, V_k$, each stable under x. On V_i, x clearly has characteristic polynomial $(T - a_i)^{m_i}$. By the CRT, we can find $p(T)$ s.t. $p(T) \equiv \begin{cases} a_i \mod (T - a_i)^{m_i} \\ 0 \mod T \end{cases}$. Set $q(T) = T - p(T)$. then each of $p(T), q(T)$ has zero constant term since $p(T) \equiv 0 \mod T$.

- 3

  Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in x, x_s, x_n commute with each other, as well as with all endomorphisms which commute with x. They also stabilise all subspaces of V stabilised by x, in particular the V_i. The congruence $p(T) \equiv a_i \mod (T - a_i)^{m_i}$ shows the restriction of $x_s - a_i \cdot 1$ to V_i is zero for all i, hence x_s acts diagonally on V_i with single eigenvalue a_i. By definition, $x_n = x - x_s$, which makes it clear that x_n is nilpotent. Because $p(T), q(T)$ have no constant term, 3 follows.

- 1

  Let $x = s + n$ be another such decomposition, so $x_s - s = n - x_n$. By 2, all endomorphisms in sight commute. Sums of commuting semisimple or nilpotent endomorphisms are again semisimple or nilpotent respectively, whereas only 0 can be both semisimple and nilpotent. This forces $\begin{cases} s = x_s \\ n = x_n \end{cases}$

$\rightarrow$ the decomposition $x = x_s + x_n$ is called the **Jordan-Chevalley decomposition of** x; x_s, x_n are called **the semisimple part** and **the nilpotent part** of x.

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, V finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is ad$x$. Similarly, if x is semisimple, then so is ad$x$.
pf. Choose a basis $(v_1, \ldots, v_n)$ of V relative to which x has matrix diag$(a_1, \ldots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \ldots, v_n)$, i.e. $e_{ij}(v_k) = \delta_{jk} v_i$. Then ad$x(e_{ij}) = (a_i - a_j)e_{ij}$, so ad$x$ has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

Lemma 3. *Let $x \in EndV$, $x = x_s + x_n$ its Jordan decomposition. Then $adx = adx_s + adx_n$ is the Jordan decomposition of adx.*

pf. We have seen adx_s, adx_n are semisimple, nilpotent respectively and they commute, given $[\mathrm{ad}x_s, \mathrm{ad}x_n] = \mathrm{ad}[x_s, x_n] = 0$. Then by the previous proposition 1, the lemma follows.

Lemma 4. *Let $\mathfrak{U}$ be a finite dimensional F-algebra. Then Der$\mathfrak{U}$ contains the semisimple and nilpotent parts of all its elements.*

pf. If $\delta \in \mathrm{Der}\mathfrak{U}$, let $\sigma, \nu \in \mathrm{End}\mathfrak{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathfrak{U}_a = \{x \in \mathfrak{U} \mid (\delta - a1)^k x = 0, \ \exists k\}$. Then $\mathfrak{U}$ is the direct sum of those $\mathfrak{U}_a$ for which a is an eigenvalue of δ and σ acts on $\mathfrak{U}_a$ as scalar multiplication by a. $\forall a, b \in F$, $\mathfrak{U}_a\mathfrak{U}_b \subset \mathfrak{U}_{a+b}$ since $(\delta - (a+b)1)^n(xy) = \sum_{i=0}^n \binom{n}{i}((\delta - a1)^{n-i}x) \cdot ((\delta - b1)^i y)$, $\forall x, y \in \mathfrak{U}$. If $x \in \mathfrak{U}_a, y \in \mathfrak{U}_b$, $\sigma(xy) = (a+b)xy$ given $xy \in \mathfrak{U}_{a+b}$. On the other hand, $(\sigma x)y + x(\sigma y) = (a+b)xy$. By the directness of the sum $\mathfrak{U} = \coprod \mathfrak{U}_a$, σ is a derivation. Since $\nu = \delta - \sigma$, $\delta, \sigma \in \mathrm{Der}\mathfrak{U}$ so is ν.

Lemma 5. *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) \mid [xB] \subset A\}$. Suppose $x \in M$ satisfies $\mathrm{T}(xy) = 0$, $\forall y \in M$. Then x is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of x. Fix a basis $v_1, \ldots, v_m$ of V relative to which s has matrix diag$(a_1, \ldots, a_m)$. Let E be the vector subspace of F over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \ldots, a_m$. Since E has finite dimension over $\mathbb{Q}$, we can use the dual space E^* consisting of any linear function $f : E \rightarrow \mathbb{Q}$. Given f, let y be the element of $\mathfrak{gl}(V)$ whose matrix relative to our given basis is diag$(f(a_1), \ldots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$, $\begin{cases} \mathrm{ad}s(e_{ij}) = (a_i - a_j)e_{ij} \\ \mathrm{ad}y(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}$. Let $r(t) \in F[t]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$, $\forall i, j$. By the Lagrange interpolation we can find such $r(t)$. Since ad$s$ is the semisimple part of ad$x$, it can be written as a polynomial in ad$x$ without constant term by the previous proposition. Thus ad$y$ is also a polynomial in ad$x$ without constant term. By hypothesis, ad$x$ maps B into A, we have ad$y(B) \subset A$, i.e. $y \in M$. Then $\mathrm{T}(xy) = 0 \rightarrow \sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$-linear combination of elements of E and by applying f, we obtain $\sum f(a_i)^2 = 0$. But the numbers $f(a_i)$ are rational, so this forces all of them to be 0. Finally, f must be identically 0 because a_i span E. So E^* is 0 and therefore $E = 0$.

If x, y, z are endomorphisms of a finite dimensional vector space,

$$\mathrm{T}([xy]z) = \mathrm{T}(x[yz]) \tag{1}$$

pf. $\begin{cases} [xy]z = xyz - yxz \\ x[yz] = xyz - xzy \end{cases}$. Since $\mathrm{T}(y(xz)) = \mathrm{T}((xz)y)$ given the identity above, $\mathrm{T}([xy]z) - \mathrm{T}(x[yz]) = \mathrm{T}(xzy) - \mathrm{T}(yxz) = \mathrm{T}(xzy) - \mathrm{T}(xzy) = 0$.

Theorem 4 (Cartan's Criterion)**.** *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. Suppose* $\mathrm{T}(xy) = 0,\ \forall x \in [LL],\ \forall y \in L$. *Then L is solvable.*

pf. Let V as above and $A = [LL],\ B = L$. Then $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$. Obviously $L \subset M$. By the hypothesis $\mathrm{T}(xy) = 0,\ \forall x \in [LL],\ \forall y \in L$. If $[xy]$ is a typical generator of $[LL]$ and if $z \in M$, by the previous equation A.1., $\mathrm{T}([xy]z) = \mathrm{T}(x[yz]) = \mathrm{T}([yz]x)$. Then by definition of M, $[yz] \in [LL]$ and therefore the RHS is 0 by hypothesis. Thus $\forall x \in [LL]$ satisfies $\mathrm{T}(xz) = 0,\ \forall z \in M$, so by the previous lemma x is nilpotent, hence $\mathrm{ad}_{[LL]}x$ is nilpotent. By Engel's theorem $[LL]$ is nilpotent, hence solvable. Since $[LL]$ is a solvable ideal and $L/[LL]$ is abelian, L is solvable.

Corollary 5. *Let L be a Lie algebra s.t.* $\mathrm{T}(adxady) = 0,\ \forall x \in [LL],\ \forall y \in L$. *Then L is solvable.*

pf. By the previous theorem to the adjoint representation of L, we get $\mathrm{ad}L$ solvable. Since $\ker \mathrm{ad} = Z(L)$ is solvable, L itself is solvable.

Definition 13. *Let L be any Lie algebra. If $x, y \in L$, define $\kappa(x,y) = \mathrm{T}(adxady)$. Then κ is a symmetric bilinear form on L, called the Killing form.*

$\rightarrow \kappa([xy], z) = \kappa(x, [yz])$, κ is associative.

pf. $\begin{cases} \kappa([xy], z) = \mathrm{T}(\mathrm{ad}[xy]\mathrm{ad}z) \\ \kappa(x, [yz]) = \mathrm{T}(\mathrm{ad}x\mathrm{ad}[yz]) \end{cases}$ are equivalent by the previous equation A.1.

Lemma 6. *Let I be an ideal of L. If κ is the killing form of L and κ_I is the killing form of I, then $\kappa_I = \kappa|_{I\times I}$.*

pf. Recall if W is a subspace of a vector space V and ϕ is an endomorphism of V mapping V into W, then $\mathrm{T}\,\phi = \mathrm{T}(\phi|_W)$. If $x, y \in I$, $\mathrm{ad}x\mathrm{ad}y$ is an endomorphism of L mapping L into I, so its trace $\kappa(x,y)$ coincides with the trace $\kappa_I(x,y)$ of $\mathrm{ad}x\mathrm{ad}y|_I = \mathrm{ad}_I x \mathrm{ad}_I y$.

A symmetric bilinear form $\beta(x,y)$ is called nondegenerate if its radical S is 0, where $S = \{x \in L \mid \beta(x,y) = 0,\ \forall y \in L\}$. Here S is an ideal of L since the killing form is associative. Since a Lie algebra is called semisimple when $\mathrm{rad}L = 0$, this is equivalent to requiring L to have no nonzero abelian ideals.

Theorem 5. *Let L be a Lie algebra. Then L is semisimple iff its killing form is nondegenerate.*

pf.

- Suppose $\mathrm{rad}L = 0$. Let S be the radical of κ. Then $\mathrm{T}(\mathrm{ad}x\mathrm{ad}y) = 0,\ \forall x \in S,\ \forall y \in L$. By the Cartan's criterion, $\mathrm{ad}_L S$ is solvable, hence S is solvable. Then since S is an ideal of L, $S \subset \mathrm{rad}L = 0$ and κ is nondegenerate.
- Let $S = 0$. Suppose $x \in I, y \in L$, where I is abelian ideal of L. Then $\mathrm{ad}x\mathrm{ad}y$ maps $L \to L \to I$ and $(\mathrm{ad}x\mathrm{ad}y)^2$ maps L into $[II] = 0$. This means $\mathrm{ad}x\mathrm{ad}y$ is nilpotent, hence $0 = \mathrm{T}(\mathrm{ad}x\mathrm{ad}y) = \kappa(x,y)$, so $I \subset S = 0$.

$\rightarrow S \subset \mathrm{rad}L$ but the converse does not necessarily hold.

A Lie algebra L is said to be the direct sum of ideals $I_1, \ldots, I_t$ provided $L = I_1 + \cdots + I_t$. This forces $[I_i I_j] \subset I_i \cap I_j = 0,\ \forall i \neq j$. Write $L = I_1 \oplus \cdots \oplus I_t$.

Theorem 6. *Let L be semisimple. Then there exists ideals $L_1, \ldots, L_t$ of L which are simple, s.t. $L = L_1 \oplus \cdots \oplus L_t$. Every simple ideal of L coincides with one of L_i. Moreover, the killing form of L_i is the restriction of κ to $L_i \times L_i$.*

pf. Let I be an arbitrary ideal of L. Then $I^\perp = \{x \in L \mid \kappa(x,y) = 0,\ \forall y \in I\}$ is also an ideal by the associativity of κ. By Cartan's criterion $I \cap I^\perp$ of L is solvable. Therefore given $\dim I + \dim I^\perp = \dim L$, $L = I \oplus I^\perp$.

- Existence

  Let $\dim L = 1$. Then the theorem follows trivially. When $\dim L > 1$, L has nonzero proper ideal, let a minimal nonzero ideal say L_1. Then $L = L_1 \oplus L_1^{\perp}$. Any ideal of L_1 is an ideal of L, L_1 is semisimple by the minimality. For the same reason, $L_1^{\perp}$ is semisimple. Then by inductive hypothesis, both split into direct sums of simple ideals, which are also ideals of L. The decomposition of L follows.

- Uniqueness

  If I is any simple ideal of L, $[IL]$ is also an ideal of I, nonzero, because $Z(L) = 0$. Thus $[IL] = I$. On the other hand, $[IL] = [IL_1] \oplus \cdots \oplus [IL_t]$ so all but one summand must be 0. Say $[IL_i] = I$. Then $I \subset L_i$ and $I = L_i$.

Corollary 6. *If L is semisimple, $L = [LL]$ and all ideals and homomorphic images of L are semisimple. Moreover each ideal of L is a sum of certain simple ideals of L.*

Theorem 7. *If L is semisimple, $adL = DerL$, i.e. every derivation of L is inner.*

pf. Since L is semisimple, $Z(L) = 0$. Therefore $L \to \mathrm{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \mathrm{ad}L$ itself has nondegenerate killing form. If $D = \mathrm{Der}L$, $[DM] \subset M$. Then by the previous lemma, κ_M is the restriction to $M \times M$ of the killing form κ_D of D. In particular, if $I = M^{\perp}$ is the subspace of D orthogonal to M under κ_D, the nondegeneracy of κ_M forces $I \cap M = 0$. Both I, M are ideals of D, so $[IM] = 0$. If $\delta \in I$, this forces $\mathrm{ad}(\delta x) = 0$, $\forall x \in L$ since $[\delta \mathrm{ad}x] = \mathrm{ad}(\delta x)$, so $\delta x = 0$ because ad is one to one and $\delta = 0$. Thus $I = 0$, $\mathrm{Der}L = M = \mathrm{ad}L$.
$\to$ $\mathrm{Der}L$ coincides with $\mathrm{ad}L$, and $L \to \mathrm{ad}L$ is one to one, and therefore each $x \in L$ determines unique elements $s, n \in L$ s.t. $\mathrm{ad}x = \mathrm{ad}s + \mathrm{ad}n$, the usual Jordan decomposition of ad x. This means $x = s + n$, $[sn] = 0$, s ad-semisimple and n ad-nilpotent.

Let L be a Lie algebra. A vector space V, endowed with an operation $\begin{array}{ccc} L \times V & \to & V \\ (x,v) & \mapsto & xv \end{array}$ is called an L-module if the following conditions are satisfied; $\begin{cases} (ax+by)v = a(xv) + b(yv) \\ x(av+bw) = a(xv) + b(xw) \\ [xy]v = xyv - yxv \end{cases}$ $\forall x, y \in L$, $\forall v, w \in V$, $a, b \in F$.
e.g. if $\phi : L \to \mathfrak{gl}(V)$ is a representation of L, then V may be viewed as an L-module via the action $xv = \phi(x)(v)$. Conversely, given an L-module V, this defines a representation $\phi : L \to \mathfrak{gl}(V)$.

A homomorphism of L-modules is a linear map $\phi : V \to W$ s.t. $\phi(xv) = x\phi(v)$. The kernel of such a homomorphism is then an L-submodule of V. When ϕ is an isomorphism of vector spaces, we call it an isomorphism of L-modules, and the two modules are said to afford equivalent representations of L. An L-module V is called irreducible if it has precisely two L-submodules. V is called completely reducible if V is a direct sum of irreducible L-submodules or equivalently if each L-submodule W of V has a complement W'. Given a representation $\phi : L \to \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in $\mathrm{End}V$ leaves invariant precisely the same subspaces as L.

Lemma 7 (Schur). *Let $\phi : L \to \mathfrak{gl}(V)$ be irreducible. Then the only endomorphisms of V commuting with all $\phi(x)$ are scalars.*

$\to$ holds for L as well by the above statement.

Let V be an L-module. Then the dual vector space V^* becomes an L-module, called **contragredient** if we define $\forall f \in V^*$, $v \in V, x \in L : (xf)(v) = -f(xv)$
$(([xy]f)(v) = -f([xy]v) = -f(xyv - yxv) = -f(xyv) + f(yxv) = (xf)(yv) - (yf)(xv) = -(yxf)(v) + (xyf)(v) = ((xy - yx)f)(v)$).

If V, W are L-modules, let $V \otimes W$ be the tensor product over F of the underlying vector spaces. For Lie algebras, $x(v\otimes w)=xv\otimes w+v\otimes xw$ ($[xy](v\otimes w)=[xy]v\otimes w+v\otimes[xy]w=(xyv-yxv)\otimes w+v\otimes(xyw-yxw)=(xyv\otimes w+v\otimes xyw)-(yxv\otimes w+v\otimes yxw)=(xy-yx)(v\otimes w)$). Given a vector space V over F, there is a standard isomorphism of vector spaces $\begin{array}{ccc} V^*\otimes V & \to & \mathrm{End}V \\ f\otimes v & \mapsto & (w\mapsto f(w)v)\end{array}$. If V is in addition an L-module, $V^*\otimes V$ becomes an L-module. Therefore End$V$ can also be viewed as an L-module. This action of L on End$V$ can be described as follows; $(xf)(v)=xf(v)-f(xv),\ \ x\in L,\ \ f\in \mathrm{End}V,\ \ v\in V$.

Let L be semisimple and $\phi: L\to \mathfrak{gl}(V)$ be a faithful representation of L. Define a symmetric bilinear form $\beta(x,y)=\mathrm{T}(\phi(x)\phi(y))$ on L. The form β is associative so its radical S is an ideal of L. Moreover β is nondegenerate. If $(x_1,\ldots,x_n)$ is a basis of L, there is a uniquely determined dual basis $(y_1,\ldots,y_n)$ relative to β satisfying $\beta(x_i,y_j)=\delta_{ij}$. If $x\in L$, we can write $\begin{cases}[xx_i]=\sum_j a_{ij}x_j\\ [xy_i]=\sum_j b_{ij}y_j\end{cases}$. By the associativity of β, $a_{ik}=\sum_j a_{ij}\beta(x_j,y_k)=\beta([xx_i],y_k)=\beta(-[x_ix],y_k)=\beta(x_i,-[xy_k])=-\sum_j b_{kj}\beta(x_i,y_j)=-b_{ki}$. If $\phi: L\to\mathfrak{gl}(V)$ is any representation of L, write $c_\phi(\beta)=\sum_i\phi(x_i)\phi(y_i)\in \mathrm{End}V$. Since $a_{ik}=-b_{ki}$, $[\phi(x)c_\phi(\beta)]=\sum_i[\phi(x)\phi(x_i)]\phi(y_i)+\sum_i\phi(x_i)[\phi(x)\phi(y_i)]=\sum_{ij}a_{ij}\phi(x_j)\phi(y_i)+\sum_{ij}b_{ij}\phi(x_i)\phi(y_j)=0$, i.e. $c_\phi(\beta)$ is an endomorphism of V commuting with $\phi(L)$.

Definition 14. *Let $\phi: L\to\mathfrak{gl}(V)$ be a faithful representation with nondegenerate trace form $\beta(x,y)=\mathrm{T}(\phi(x)\phi(y))$. Then having fixed a basis $(x_1,\ldots,x_n)$ of L, the Casimir element of ϕ is $c_\phi(\beta)$. Its trace is $\sum_i\mathrm{T}(\phi(x_i)\phi(y_i))=\sum\beta(x_i,y_i)=\dim L$.*

$\to \phi$ is irreducible, then by the Schur's lemma, c_ϕ is a scalar and it is independent of the bases of L.

e.g. $L=\mathfrak{sl}(2,F), V=F^2, \phi: L\to\mathfrak{gl}(V)$ the identity map. Let (x,h,y) be the standard basis of L. Then the dual basis relative to the trace form is $(y,h/2,x)$ so $c_\phi=xy+(1/2)h^2+yx=\begin{bmatrix}3/2 & 0\\ 0 & 3/2\end{bmatrix}$ where $3/2=\dim L/\dim V$.

When ϕ is not faithful, $\ker\phi$ is still an ideal of L, hence a sum of certain simple ideals by the previous corollary. Let L' denote the sum of the remaining simple ideals in the previous theorem. Then the restriction of ϕ to L' is a faithful representation of L' and we make the preceding construction using the dual basis of L'. The resulting element of End$V$ is called the Casimir element of ϕ, denoted by c_ϕ.

Lemma 8. *Let $\phi: L\to\mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L. Then $\phi(L)\subset\mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L-module.*

pf. Since $L=[LL]$ and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$ the lemma follows.

Theorem 8 (Weyl)**.** *Let $\phi: L\to\mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then ϕ is completely reducible.*

pf.

- V has an L-submodule W of codimension 1.

  Since L acts trivially on V/W, by the previous lemma, we may denote this module F without confusion and there exists a s.e.s. $0\to W\to V\to F\to 0$. Using inductive hypothesis, we can reduce to the case where W is an irreducible L-module as follows; let W' be a proper nonzero submodule of W. Then this yields a s.e.s. $0\to W/W'\to V/W'\to F\to 0$. By inductive hypothesis, this splits i.e. there exists an one-dimensional L-submodule of V/W', say $\tilde{W}/W'$, complementary to W/W'. So there is another s.e.s. $0\to W'\to\tilde{W}\to F\to 0$ where $\dim W'<\dim W$. So by the inductive hypothesis, submodule X complementary to W' in $\tilde{W}=W'\oplus X$. But $V/W'=W/W'\oplus\tilde{W}/W'$. So $V=W\oplus X$, since the dimensions add up to $\dim V$ and $W\cap X=0$. So we may assume W is irreducible. Let $c=c_\phi$ be the Casimir element of ϕ. Since c commutes with $\phi(L)$, c is actually an L-module endomorphism of V and in particular $c(W)\subset W$ and $\ker c$ is an L-submodule of V. Since L acts trivially on V/W i.e. $\phi(L)$ sends V into W, c must do likewise. So c has trace 0 on V/W. On the other hand, c acts as a scalar on the irreducible L-submodule W by the Schur's lemma, and this cannot be 0, because that would force $\mathrm{T}_V(c) = 0$, a contradiction. So here, c acts invertibly on W (nonzero scalar) so $\ker c \cap W = 0$, and c maps $V \to W$ so $\dim \ker c \geq \dim V - \dim W = 1$; combined with $\ker c \cap W = 0$ that gives $\dim \ker c = 1$. Thus $\ker c$ is an one-dimensional L-submodule of V which intersects W trivially, a desired complement to W.

- Let W be a nonzero submodule of V. Then there exists a s.e.s. $0 \to W \to V \to V/W \to 0$. Let $\hom(V, W)$ be the space of linear maps $V \to W$ viewed as L-module. Let $\mathcal{V}$ be the subspace of $\hom(V, W)$ consisting of those maps whose restriction to W is a scalar multiplication. $\mathcal{V}$ is actually an L submodule ($f|_W = a1_W \to \forall x \in L, \forall w \in W,\ (xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0$, so $xf|_W = 0$). Let $\mathcal{W}$ be the subspace of $\mathcal{V}$ consisting of those f whose restriction to W is zero. Then $\mathcal{W}$ is also an L-submodule and L maps $\mathcal{V}$ into $\mathcal{W}$. Also $\mathcal{V}/\mathcal{W}$ has dimension 1, because each $f \in \mathcal{V}$ is determined modulo $\mathcal{W}$ by the scalar $f|_W$. This gives us a s.e.s $0 \to \mathcal{W} \to \mathcal{V} \to F \to 0$. Then by the first case, $\mathcal{V}$ has an one-dimensional submodule complementary to $\mathcal{W}$. Let $f : V \to W$ span it, so after multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. $0 = (xf)(v) = xf(v) - f(xv)$, i.e. f is an L-homomorphism, so L kills f and $\ker f$ is an L-submodule of V. Since f maps V into W and acts as 1_W on W, $V = W \oplus \ker f$.

Theorem 9. *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra. Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decomposition in L coincide.*

pf. Let $x \in L$ be arbitrary with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\mathrm{ad}x(L) \subset L$, the $\mathrm{ad}x_s(L) \subset L$ and $\mathrm{ad}x_n(L) \subset L$ where ad$= ad_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including L as an ideal. If W is any L-submodule of V, define $L_W = \{y \in \mathfrak{gl}(V) \mid y(W) \subset W,\ \mathrm{T}(y|_W) = 0\}$. Since $L = [LL]$, L lies in all such L_W. Set L' = intersection of N with all spaces L_W. Clearly L' is a subalgebra of N including L as an ideal. Even more, if $x \in L$, $x_s, x_n \in L_W \subset L'$. By the Weyl's theorem, we can write $L' = L + M$ for some L-submodule M where the sum is direct. But $[LL'] \subset L$ given $L' \subset N$, so the action of L on M is trivial. Let W be any irreducible L-submodule of V. If $y \in M$, $[Ly] = 0$ so by the Schur's lemma y acts on W as a scalar. On the other hand $\mathrm{T}(y|_W) = 0$ because $y \in L_W$. Thus y acts on W as zero. V can be written as a direct sum of irreducible L-submodules by Weyl's theorem so $y = 0$. This means $M = 0$, $L = L'$.

Corollary 7. *Let L be a semisimple Lie algebra, $\phi : L \to \mathfrak{gl}(V)$ a representation of L. If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. The algebra $\phi(L)$ is spanned by the eigenvectors of $ad_{\phi(L)}\phi(s)$ since L has this property relative to $\mathrm{ad}s$. Thus $\mathrm{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\mathrm{ad}_{\phi(L)}\phi(n)$ is nilpotent, and it commutes with $\mathrm{ad}_{\phi(L)}\phi(s)$. Accordingly, $\phi(x) = \phi(s) + \phi(n)$ is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

e.g.

- **the representations of $\mathfrak{sl}(2, F)$**

  Let L denote $\mathfrak{sl}(2, F)$ whose standard basis consists of $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $[hx] = 2x, [hy] = -2y, [xy] = h$. Let V be an arbitrary L-module. Since h is semisimple, by the previous corollary h acts diagonally on V. This yields a decomposition of V as direct sum of eigenspaces $V_\lambda = \{v \in V \mid hv = \lambda v\}$, $\lambda \in F$, given F algebraically closed. Whenever $V_\lambda \neq 0$, λ is called a **weight** of h in V and V_λ is called a **weight space**.

  Lemma 9. *If $v \in V_\lambda$, $xv \in V_{\lambda+2}$, $yv \in V_{\lambda-2}$.*

  pf. $h(xv) = [hx]v + xhv = 2xv + \lambda xv = (\lambda + 2)xv$ and similarly for y.
  $\to x, y$ are represented by nilpotent endomorphisms of V.

Since $\dim V < \infty$, $V = \coprod_{\lambda \in F} V_\lambda$ is direct, there must exists $V_\lambda \neq 0$ s.t. $V_{\lambda+2} = 0$ and by the previous lemma $xv = 0$, $\forall v \in V_\lambda$. For such λ, any nonzero vector in V_λ is called a **maximal vector** of weight λ.

Assume V is an irreducible L-module. Choose a maximal vector $v_0 \in V_\lambda$. Set $v_{-1} = 0$, $v_i = (\frac{1}{i!})y^i v_0$, $i \geq 0$.

Lemma 10. $\begin{cases} hv_i = (\lambda - 2i)v_i \\ yv_i = (i+1)v_{i+1} \\ xv_i = (\lambda - i + 1)v_{i-1}, i \geq 0 \end{cases}$

pf.

- by the previous lemma
- by the definition, $(i+1)v_{i+1} = (i+1)\frac{1}{(i+1)!}y^{i+1}v_0 = \frac{1}{i!}y^{i+1}v_0 = yv_i$.
- When $i = 0$, $xv_0 = (\lambda - i + 1)v_{-1} = 0$ holds trivially. When $i > 0$, $\begin{aligned} ixv_i &= xyv_{i-1} && \text{b} \\ &= [xy]v_{i-1} + yxv_{i-1} = hv_{i-1} + yxv_{i-1} \\ &= (\lambda - 2(i-1))v_{i-1} + (\lambda - i + 2)yv_{i-2} && \text{b} \\ &= (\lambda - 2i + 2)v_{i-1} + (i-1)(\lambda - i + 2)v_{i-1} && \text{b} \\ &= i(\lambda - i + 1)v_{i-1} \end{aligned}$

  $xv_i = (\lambda - i + 1)v_{i-1}$ by dividing both sides by i.

$\rightarrow$ the nonzero v_i is linearly independent by the first identity of the lemma. Let m be the smallest integer for which $v_m \neq 0$, $v_{m+1} = 0$. Then by the previous lemma, the subspace of V with basis $(v_0, \ldots, v_m)$ is an L-submodule different from 0. Because V is irreducible, this must be all of V.
$\rightarrow$ for $i = m+1$, the third identity's LHS becomes 0, whereas the RHS is $(\lambda - m)v_m$. Since $v_m \neq 0$, $\lambda = m$, i.e. the weight of a maximal vector is a nonnegative integer. We call it **the highest weight** of V and each weight μ occurs with multiplicity 1 by the first identity. Since V determines λ uniquely, the maximal vector v_0 is the only possible one in V.

Theorem 10. *Let V be an irreducible module for $L = \mathfrak{sl}(2, F)$.*

1. *Relative to h, V is the direct sum of weight spaces V_μ, $\mu = m, m-2, \ldots, -(m-2), -m$ where $m + 1 = \dim V$ and $\dim V_\mu = 1$ for each μ.*
2. *V has a unique maximal vector, whose weight is m.*
3. *The action of L on V is given explicitly by the above formulas if the basis is chosen in the prescribed fashion. In particular, there exists at most one irreducible L-module upto isomorphism of each possible dimension $m+1$, $m \geq 0$.*

pf.

- Relative to the ordered basis $(v_0, \ldots, v_m)$, the matrices of the endomorphisms representing x, y, h can be written explicitly and h yields a diagonal matrix, while x, y yield upper and lower triangular nilpotent matrices respectively. For $i = m+1$ with the previous lemma's third identity gives $(\lambda - m)v_m = 0$ but since $v_m \neq 0$, $\lambda = m$, i.e. the weight of a maximal vector is a nonnegative integer. Moreover each weight occurs with multiplicity one by the previous lemma's first identity, $\dim V_\mu = 1$ if $V_\mu \neq 0$.
- In the previous statement since V determines λ uniquely, the maximal vector v_0 is the only possible one in V.
- by the statement after the lemma

Corollary 8. *Let V be any finite dimensional L-module, $L = \mathfrak{sl}(2, F)$. Then the eigenvalues of h on V are all integers, and each occurs along with its negative. Moreover, in any decomposition of V into direct sum of irreducible submodules, the number of summands is precisely* $\dim V_0 + \dim V_1$.

pf. If $V = 0$, there is nothing to prove. Otherwise, by the Weyl's theorem, we can write V as a direct sum of irreducible submodules. The latter are described by the previous theorem, so the first assertion of the corollary is obvious. For the second, each irreducible L-module has a unique occurrence of either the weight 0 or else the weight 1.

For the $m+1$-dimensional vector space over F with basis $(v_0, \dots, v_m)$, it is called $V(m)$. Let $\phi : L \to \mathfrak{gl}(V(m))$ be the irreducible representation of highest weight m. Then $\phi(x), \phi(y)$ are nilpotent endomorphisms, so we can define an automorphism of $V(m)$ by $\tau = \exp \phi(x) \exp \phi(-y) \exp \phi(x)$. Assume wlog $m > 0$ so that the representation is faithful, given L being simple. Then conjugating $\phi(h)$ by τ has the same effect as applying $\exp(\mathrm{ad}\phi(x)) \exp(\mathrm{ad}\phi(-y)) \exp(\mathrm{ad}\phi(x))$ to $\phi(h)$. But $\phi(L) \cong L$ so $\tau\phi(h)\tau^{-1} = -\phi(h)$ or $\tau\phi(h) = -\phi(h)\tau$. So τ sends the basis vector v_i of weight $m - 2i$ to the basis vector v_{m-i} of weight $-(m - 2i)$, the symmetry in the structure of $V(m)$.